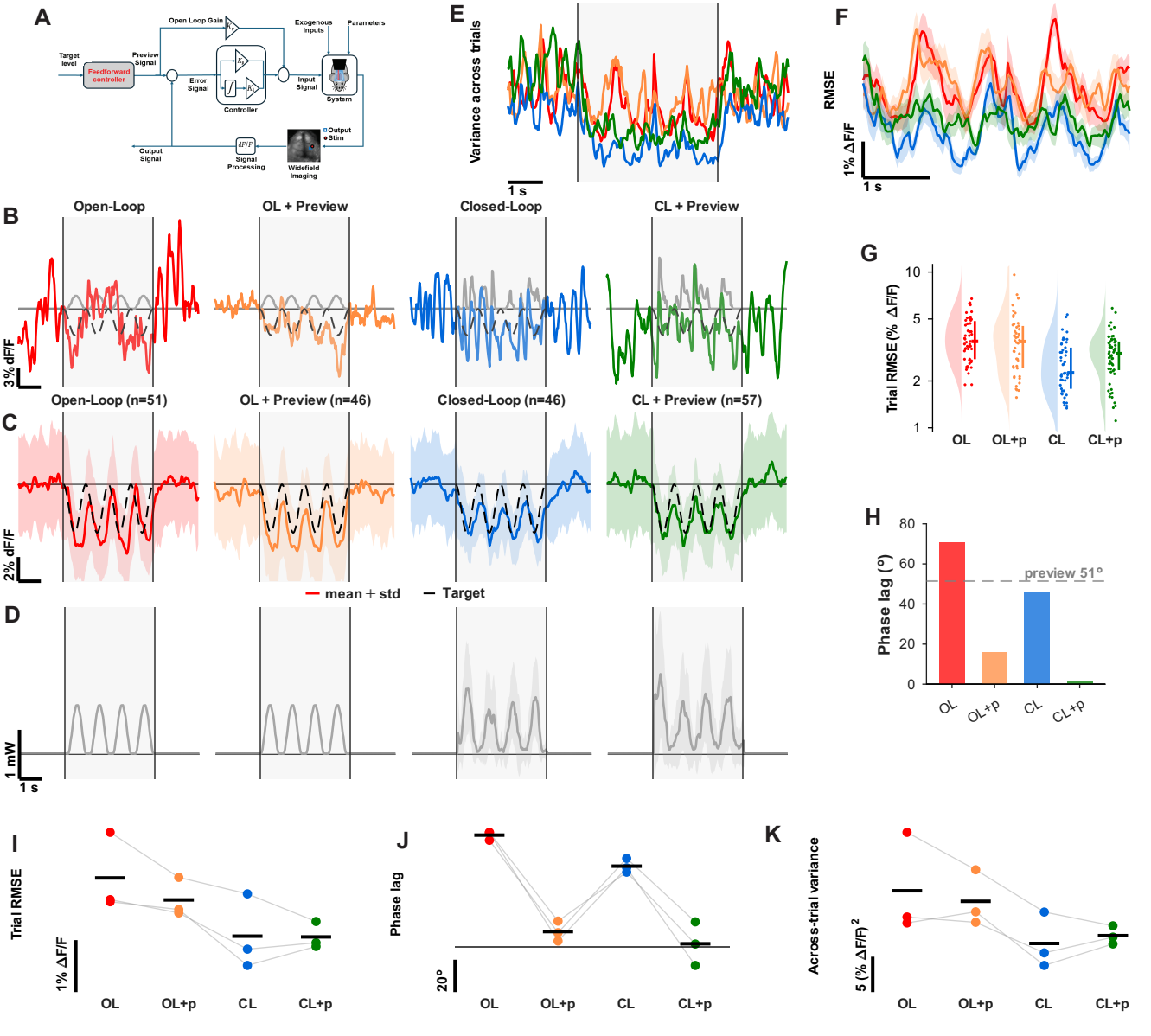

A
Open Loop Gain
Exogenous Inputs
Parameters
Target level
Preview Signal
Feedforward controller
Error Signal
Controller
Input Signal
System
Output
Stim
Output Signal
Signal Processing
Widefield Imaging
E
Variance across trials
1 s
F
RMSE
1% ΔF/F
1 s
B
Open-Loop
OL + Preview
Closed-Loop
CL + Preview
3% dF/F
G
Trial RMSE (% ΔF/F)
10
5
2
1
OL
OL+p
CL
CL+p
C
Open-Loop (n=51)
OL + Preview (n=46)
Closed-Loop (n=46)
CL + Preview (n=57)
2% dF/F
mean ± std
Target
H
Phase lag (°)
80
60
40
20
0
preview 51°
OL
OL+p
CL
CL+p
D
1 mW
1 s
I
Trial RMSE
1% ΔF/F
OL
OL+p
CL
CL+p
J
Phase lag
20°
OL
OL+p
CL
CL+p
K
Across-trial variance
5 (% ΔF/F)²
OL
OL+p
CL
CL+p